Opinion Network Formation

A similarity network built from a class survey on
Technology, Education, Ethics, and Environment

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DPCN, Assignment 1
Team Deadlines Passing, Care Not
https://github.com/avi1o1/DPCN_A1

Overview

Task We build a network from a class survey that covers four topics and then study that network.

Approach Each respondent is one node. We put an edge between two people when their survey answers are similar. We then study the network in three ways.

- *Structure* We measure degree, clustering, components, four centralities, and Louvain communities.
- *Dynamics* We run the DeGroot opinion average and an SIS contagion model from the course.
- *Null models* We compare the network with an Erdős–Rényi graph, a degree-preserving rewiring, and a label permutation. This helps us check whether the structure is more than chance.

Headline 51 of the 87 respondents, which is 59%, fall into one connected group. The other 36 do not have any strong-similarity partner.

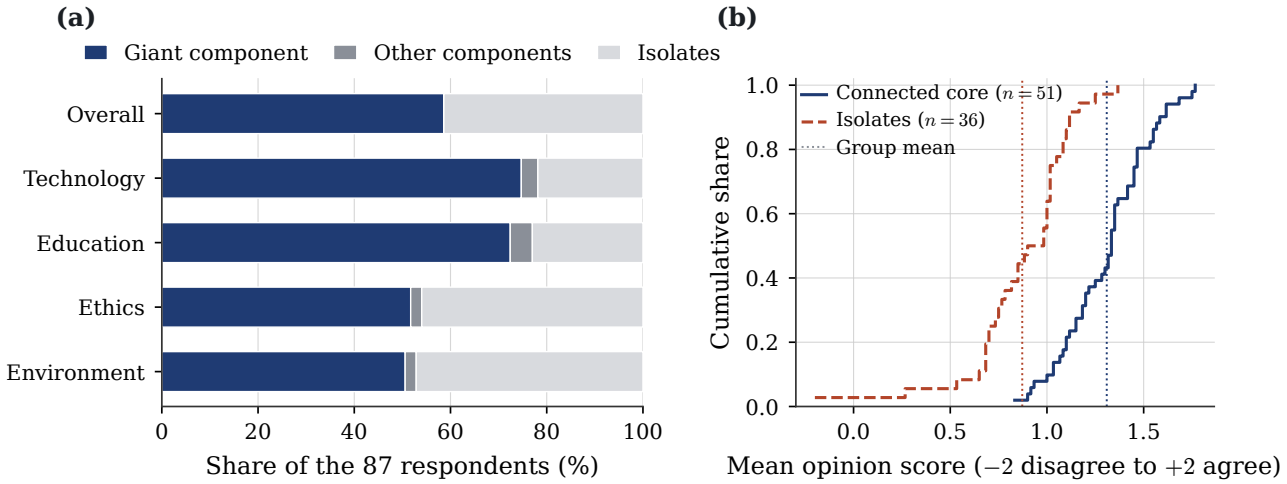


Figure 1. **(a)** How the 87 respondents divide into the giant component, the remaining small components, and the isolates, for each of the five networks. Technology and Education hold together at 75% and 72% in the giant component. Ethics and Environment fall apart at 52% and 51%. **(b)** Empirical CDF of the mean opinion score, with the dotted rules at the two group means. The respondents with no edge are the moderates. They average 0.87 against 1.31 for the connected core.

1. Dataset Documentation

The raw survey The survey has 96 responses to 60 opinion statements. Each topic has 15 statements. The topic codes are **T** for Technology, **E** for Education, **S** for Ethics, and **V** for Environment. Each statement uses a five-point Likert scale along with a "No Comments" option.

Preparation

- *Likert to number* We convert Strongly Disagree to −2 and Strongly Agree to +2. The scale is symmetric, so 0 really means neutral. The mean score then shows both direction and strength.

– *"No Comments" means missing* We do not treat a skipped answer as neutral. There are 62 blanks out of $87 \times 60 = 5220$ values, which is 1.19%. These blanks come from 19 respondents.
– *Quality filter* We remove any respondent who answered fewer than half of the 60 questions. This removes 9 responses and leaves **87 respondents**.
– *Result* We get an 87×60 table of scores in $[-2, +2]$. Each "No Comments" answer stays blank. Each question also keeps its topic and its text.

Missing values We handle blanks in two ways depending on the calculation. When a result has to match the graph exactly, we replace a blank with 0 before averaging because the edge-building step also uses that rule. For descriptive statistics, we leave the blank out of that person's average. In practice the difference is very small. The class mean changes by at most 0.01. The biggest change for any one respondent is 0.23. We keep these two uses separate and do not compare numbers that come from different rules.

2. Pipeline Followed

Similarity Each respondent becomes a vector of 60 answers, $\mathbf{x}_i \in [-2, 2]^{60}$, with blanks set to 0. We measure similarity between two people using the cosine of the angle between their vectors.

$$s_{ij} = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \, \|\mathbf{x}_j\|} \in [-1, 1].$$

Cosine similarity rewards a similar answer pattern across all 60 questions. It does not reward only a similar average score.

Edges We compute the full 87×87 cosine matrix and keep the top **10%** of pairs as edges. This is the 90th percentile cut.

– We need a hard cut. Most people in this class agree on average, so a loose threshold would give us an almost complete graph with very little use.
– This rule keeps a fixed *rank* and not a fixed score. So it always gives exactly **375 edges**, whatever the similarity values look like.
– We build **five networks** with the same rule. One *overall* network uses all 60 questions. Four others use one topic each, with 15 questions each.
– Node count, edge count, and density stay the same across all five networks. So any difference between them comes from the wiring pattern. Section 3.5 checks the cut itself.

3. Analysis and Visualizations

3.1 Structure

Definitions *Degree* k is the number of edges attached to a node. *Density* is the fraction of possible edges that are present. *Clustering* measures how often two neighbours of a node are also linked to each other, on a scale from 0 to 1. A *component* is a set of nodes that can all reach one another. The largest one is the *giant component*. *Modularity* Q tells us how strongly the network splits into groups compared with a random graph that keeps the same degrees.

Results

Metric	Overall	Technology	Education	Ethics	Environment
Nodes / Edges	87 / 375	87 / 375	87 / 375	87 / 375	87 / 375
Density	0.100	0.100	0.100	0.100	0.100
Avg. degree $\langle k \rangle$	8.62	8.62	8.62	8.62	8.62
Avg. clustering	0.408	0.387	0.396	0.363	0.375
Components	37	21	23	42	43
Giant component	58.6%	74.7%	72.4%	51.7%	50.6%
Giant diameter	4	5	6	4	5
Giant avg. path	1.88	2.20	2.24	1.78	1.93
Modularity Q	0.108	0.242	0.263	0.151	0.079

Table 1. Structural metrics for each network. Node count, edge count, and density are the same by construction. So the other differences come from the opinion pattern.

Analysis

- The overall network is sparse but locally dense. Density is 0.10 and clustering is 0.41. So connected people often sit in tight triangles.
- **36 of 87 respondents are isolates.** This means they do not have any strong-similarity partner in the class.
- The giant component is compact. Two connected people are 1.88 steps apart on average, and the diameter is 4.
- The degree distribution has a long tail. Respondent 114 has $k = 35$ while the mean is only 8.6. Very few people come close to that value, as Figure 2 shows.

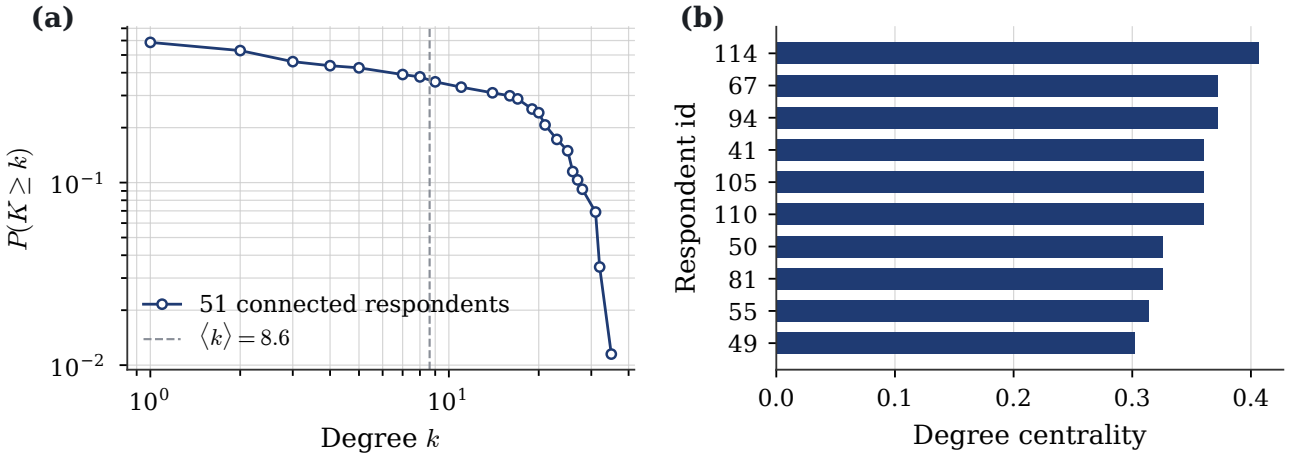


Figure 2. **(a)** Degree distribution of the overall network on log-log axes, drawn as a complementary CDF $P(K \geq k)$. Only the 51 connected respondents appear, because the 36 isolates have $k = 0$ and a log axis cannot show them. The dashed rule marks the mean degree, which sits well above the median. **(b)** The ten most central respondents by degree centrality, from 0.407 for respondent 114 down to 0.302 for respondent 49.

3.2 Centrality and communities

Definitions *Degree* centrality counts direct connections. *Closeness* measures how short the average path from one node is to the rest. *Betweenness* counts how often a node lies on a shortest path between two others, so it highlights bridge positions. *Eigenvector* centrality gives more weight to a node that is connected to other important nodes. *Louvain* looks for the grouping with the highest modularity.

Analysis

- Degree, closeness, and eigenvector centrality mostly pick the same five people, which are 114, 67, 94, 105, and 110. All five score above the class mean on every topic. They are the people who agree strongly and consistently, so their 60-value vectors look similar to many others.
- Betweenness gives a different result. Its top node is **41**, who has the maximum Technology score but only an average Education score. This mixed profile helps that person connect sub-groups.
- The most connected person and the best bridge are not the same. So one centrality measure alone is not enough here.
- Louvain finds many small communities and the modularity is low at $Q = 0.108$. Every community has a positive mean on every topic. Only one group, C6, goes below the class mean anywhere, and that happens only on Technology. These are different levels of agreement and not opposing camps.

3.3 Cohesion tracks opinion variance, not opinion strength

Observation The per-topic network results do not follow the raw averages.

Analysis

- Ethics (+1.28) and Environment (+1.37) have the highest average agreement, but their networks are the most broken at 52% and 51%.
- Technology (+0.99) and Education (+0.91) have lower average agreement, but they stay the best connected at 75% and 72%.
- The main reason is the spread. Cosine similarity depends on the answer pattern across the 15

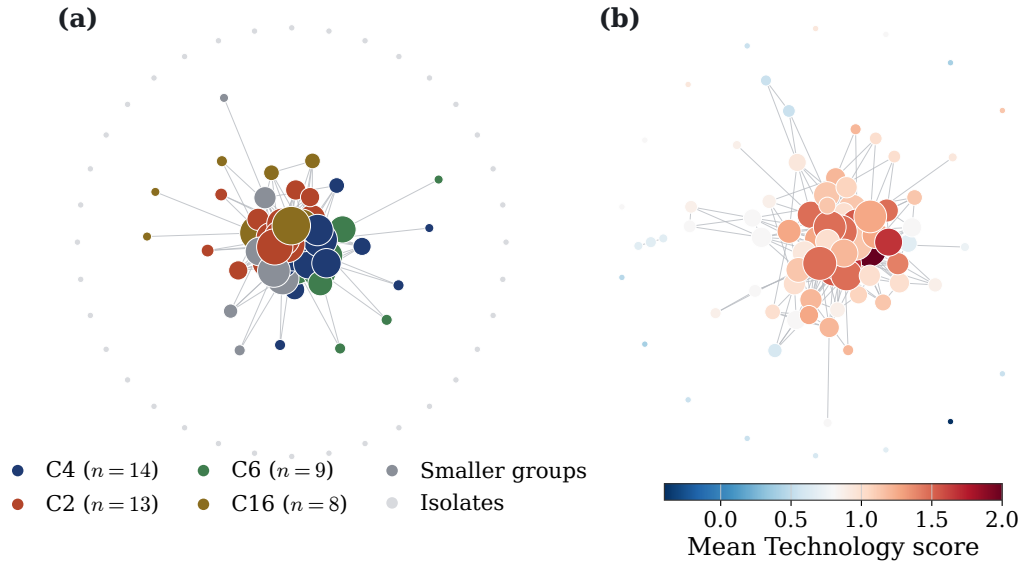


Figure 3. Force-directed node-link views with node size scaled by degree. The giant component is laid out in the centre, the remaining small components sit on an inner ring, and the isolates sit on the outer ring. **(a)** Overall network, coloured by Louvain community. The four largest communities are coloured, while singletons and pairs are grey. **(b)** Technology network, coloured by mean Technology score. People with strong agreement, in red, stay near the centre. More lukewarm answerers, in blue, stay near the edge.

Topic	Class mean	Std. dev.	Giant component	Components
Technology	0.99	0.37	74.7%	21
Education	0.91	0.35	72.4%	23
Ethics	1.28	0.45	51.7%	42
Environment	1.37	0.47	50.6%	43

Table 2. The topics with the strongest average agreement still have the most broken networks. The spread explains this better than the mean does.

questions in a topic and not just on the average level.

- Technology and Education have tightly grouped opinions with standard deviations of 0.37 and 0.35. So most vectors remain similar and the graph stays together. Ethics and Environment have stronger means but they are more spread out, with standard deviations of 0.45 and 0.47. So the same cut breaks them more.
- So the raw average can say that the class agrees more on Environment than on Technology, but the network shows a different pattern.

3.4 Dynamics on the network

DeGroot We start each respondent with their mean opinion score. At each step, every respondent replaces that value with the average of their neighbours' opinions, and everybody updates at the same time. The rule is $\mathbf{x}(t{+}1) = D^{-1}A\,\mathbf{x}(t)$. Here $\mathbf{x}(t)$ is the vector of opinion values of all respondents at step t. A is the adjacency matrix of the network. D is the diagonal degree matrix, so each diagonal entry is the degree of one node. The matrix $D^{-1}A$ therefore gives the neighbour average update. The eigenvalues of this update matrix decide what happens next. From the course, the map is stable when the relevant eigenvalue has magnitude below 1. The *SLEM*, which is the second-largest eigenvalue modulus, controls the speed. The algebraic connectivity is $\lambda_2 = 1 - \mu_2$.

Results

Analysis

- The overall network has 37 components, so there is no single class consensus. The 51-person core reaches **1.409** in **23 steps**.
- The 36 isolates never change. An isolate has no neighbour to average with, so its opinion at step 1000 is the same as at step 0.
- Convergence speed follows algebraic connectivity exactly across all five networks. This matches the

Network	λ_2	SLEM	Steps	Core size	Consensus
Overall	0.627	0.397	23	51	1.409
Ethics	0.455	0.545	31	45	1.350
Technology	0.318	0.682	53	65	1.313
Education	0.264	0.736	61	63	1.275
Environment	0.229	0.771	77	44	1.349

Table 3. DeGroot consensus. Every SLEM is below 1, so every map is stable. The order of λ_2 matches the measured order of steps in every case.

stability rule from the course on real survey data.

– Edge weights make almost no difference. The weighted consensus is 1.4099 and the unweighted one is 1.4085. Row normalisation shrinks the effect of the kept similarities because they already lie in the narrow range from 0.85 to 0.95.

SIS We model a strong opinion like a contagion. An infected person can infect each neighbour at rate β and recover at rate γ. From the course, the critical ratio is $(\beta/\gamma)_c = 1/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue of the adjacency matrix. A lower threshold means the opinion spreads more easily. This calculation uses A itself and not the row-normalised $D^{-1}A$, because one infected person can infect neighbours separately instead of averaging with them.

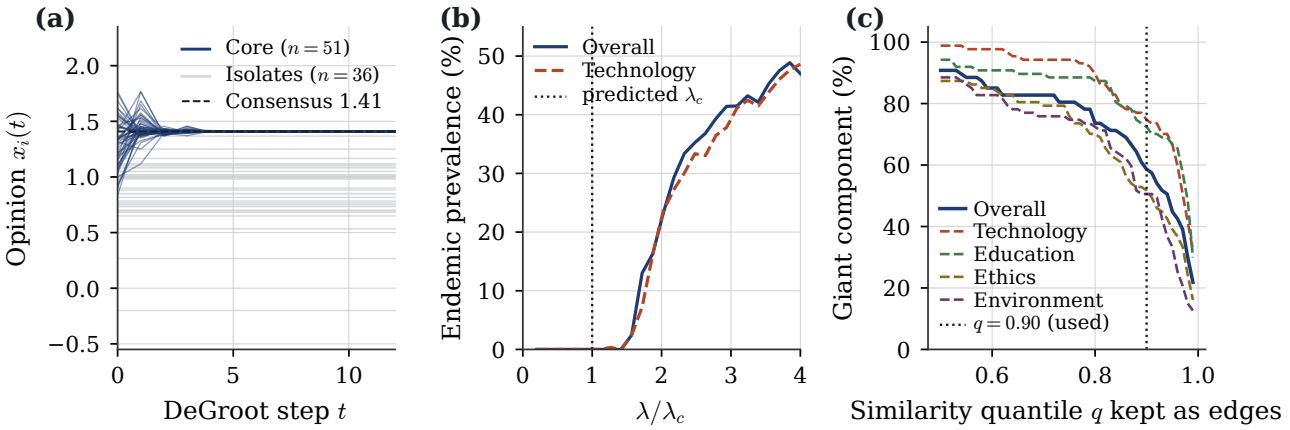


Figure 4. **(a)** DeGroot trajectories $x_i(t)$ on the overall network. The connected core settles near 1.41 while the 36 isolate lines stay flat. **(b)** Simulated SIS prevalence against λ/λ_c. The transition starts at about 1.6× the predicted bound, which is expected because the mean-field value is a lower bound and the core has only 44 to 65 people. **(c)** Giant-component share under the threshold sensitivity sweep, with the $q = 0.90$ cut used in this report marked.

3.5 Does the 10% cut-off matter?

Experiment We apply the same rule at every quantile from 0.50 to 0.99 and record the structure each time.

Results

Quantile kept	Edges	Isolates	Giant component	Clustering
0.90 (used)	375	36	58.6%	0.408
0.80	749	21	73.6%	0.579
0.70	1123	15	82.8%	0.643
0.60	1497	13	85.1%	0.674
0.50	1871	8	90.8%	0.751

Table 4. A looser cut connects most of the isolates but never all of them. At the same time it increases clustering until the graph stops being very useful.

Analysis

– The topic order stays stable at every cut. Technology always stays above Environment and Education always stays above Ethics. So this order comes from the data and not from the 90th percentile

choice.

- The isolates are only partly caused by the cut. Their count falls from 36 at $q = 0.90$ to 8 at $q = 0.50$, but it never becomes zero. So eight people remain below the median similarity with everyone else in the class.
- A looser cut also has a cost. At $q = 0.50$ the clustering rises to 0.75 and the graph becomes half complete. Then community detection and centrality do not tell us much.
- So $q = 0.90$ is a tested compromise and not just an assumption.

4. Results and Discussion

What the network says about class opinions This class is broadly agreeable. The mean opinion is positive on all four topics and the raw data does not show opposing camps. The network still reveals structure that the averages do not show clearly.

Finding 1. No global consensus and the drop-outs are the moderates

- Under DeGroot the 51-person core agrees at 1.409 while the 36 isolates never move.
- The core started at a mean of 1.309. The isolates started at 0.872.
- So the process removes the more moderate half of the class. This is a structural effect of the construction, not something the averaging model causes.

Finding 2. Lower variance gives stronger cohesion Technology and Education look better connected than Ethics and Environment even though their mean scores are lower. This happens because their opinions are less spread out. Raw agreement and social cohesion are not the same thing.

Finding 3. More fragmented topics have lower SIS thresholds

Network	$\lambda_{\max}$	$1/\lambda_{\max}$	Giant size	Giant density
Environment	23.92	0.0418	44	0.395
Ethics	22.45	0.0445	45	0.378
Overall	22.06	0.0453	51	0.294
Technology	18.17	0.0550	65	0.179
Education	17.74	0.0564	63	0.191

Table 5. Epidemic thresholds. This order is the reverse of the connectivity order in Table 1. The most broken topic spreads most easily.

- Environment and Ethics have the lowest thresholds at 0.0418 and 0.0445, yet they are the least connected topics.
- The edge count is fixed at 375, so a smaller giant component packs almost the same edges into fewer people. Environment puts 374 edges into 44 respondents. Technology spreads 373 across 65.
- A denser core raises $\lambda_{\max}$, which lowers the threshold. The standard bound $\langle k \rangle \leq \lambda_{\max} \leq k_{\max}$ applied inside the giant component predicts this ordering directly.
- Fragmentation and easy spread are not opposites here. This is a real property of the data and also a warning about the fixed-edge method.

Finding 4. The structure sits in the degree sequence A null model asks whether a random network with no opinion structure would give the same result. The z-score is $z = (\text{real} - \text{null mean})/\text{null s.d.}$. When $|z| < 2$, the real value looks close to random.

Statistic	Real	ER mean	z	Rewired mean	z
Average clustering	0.408	0.098	+27.2	0.391	+1.6
Giant fraction	0.586	1.000	−361.7	0.586	0.0
Average path length	1.880	2.292	−9.5	1.834	+3.7
Modularity	0.105	0.282	−13.7	0.100	+0.9

Table 6. The real network is compared with 500 Erdős–Rényi graphs of the same density and 500 degree-preserving rewirings of itself. Green marks a z-score inside the random band.

- Against 500 ER graphs the network is unusual on every measure, with a clustering z-score of +27

and a giant-fraction z-score of −362.

- Against 500 degree-preserving rewirings it is ordinary. Clustering, giant component, and modularity all fall inside $|z| < 2$.
- So once each person's edge count is fixed, chance supplies the rest of the structure.
- For this construction that is the same statement as "the connected core is the group that agrees with everything". Agreement level sets the degree, and after the degrees are fixed nothing is left to explain. This gives the reason for the structural analysis rather than weakening it.

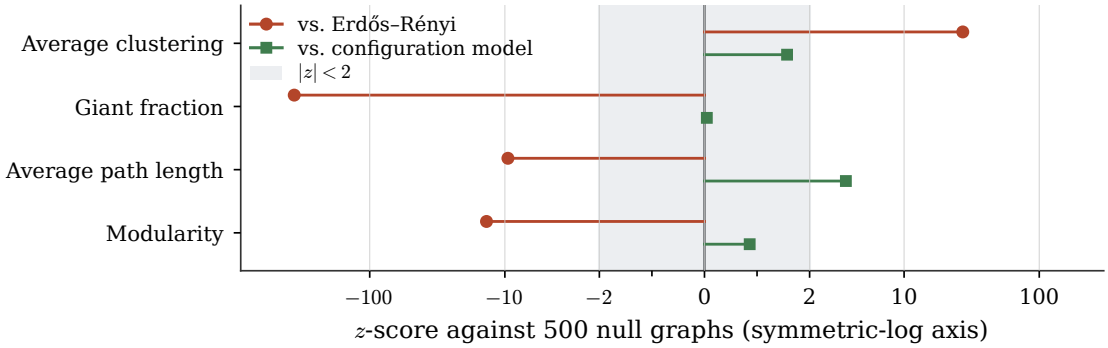


Figure 5. Null-model z-scores on a symmetric-log axis. The shaded band shows the $|z| < 2$ range that looks random. Erdős–Rényi fails on every measure. The configuration model matches clustering, giant component, and modularity.

Finding 5. The communities are real, but only at the fine level A permutation test shuffles the community labels 10,000 times while keeping the group sizes fixed. It then measures the opinion variance inside the groups. The p-value is the fraction of shuffles that do better than the real grouping.

Features tested	Observed	Null mean	z	p
All 60 question scores	0.4116	0.4592	−6.57	< 0.0001
4 topic means	0.0791	0.0839	−1.26	0.108

Table 7. The same communities pass when we test the answers used to build the graph and fail when we test only the coarser topic averages.

- On the 60 question scores the Louvain grouping is far tighter than chance ($z = -6.57$, $p < 0.0001$).
- On the 4 topic means it is not distinguishable from random ($z = -1.26$, $p = 0.108$).
- Louvain groups people by their detailed answer pattern, not by their broad position on the four topics. Two people in one community can differ on topic averages while answering the individual questions alike.
- The community-by-topic comparison in Figure 6 therefore *describes* each group but does not *define* it, and is not presented as a definition.

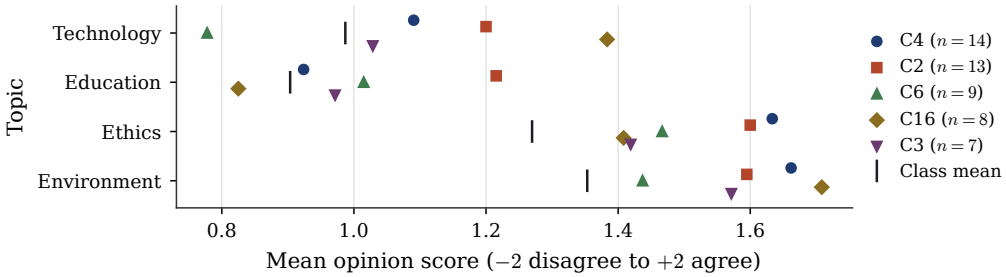


Figure 6. Mean score of each large community for each topic, against the class mean. Every value is positive and every community sits on the agreeing side of the scale, so the communities differ in level of agreement and not in direction.

In summary

- The class is agreeable across all four topics, with no real factions.

- Low variance drives the apparent cohesion, so Technology and Education out-connect Ethics and Environment even with lower means.
- The connected core is a behaviour type, the uniform strong-agreement answerers, not a demographic or a topic camp.
- Centrality measures disagree about who matters, and that disagreement is itself a result.
- Ethics and Environment move together at a respondent-level $r = 0.70$ and form one values axis. Technology and Education are nearly independent at $r = 0.29$.

5. Individual Contribution

Member	Tasks completed
Aviral	Cleaned and interpreted the survey data. Built all five networks with cosine similarity and the top-10% cut. Ran the structural analysis for degree, clustering, components, the four centralities, and the Louvain communities. Produced the static figures. Found the topic-variance result and the centrality result.
Varun	Ran the dynamics and the spectral analysis. Ran the DeGroot average with the Laplacian stability analysis. Computed the SIS epidemic threshold and its simulation. Built the Erdős–Rényi and configuration null models. Ran the community permutation test and the edge-threshold sensitivity sweep.
Arihant	Built the self-contained interactive dashboard with hand-drawn charts, a node-link view, all five networks, community and opinion colour, an isolate toggle, a respondent lookup, and the dynamics and null-model panels. Exported the dashboard data. Wrote this report and produced its figures.

Table 8. Individual contributions. The commit history supports this split.